

PART 9

THE DATA OF

MACROECONOMICS

20 MEASURING A NATION'S WELL-BEING AND THE PRICE LEVEL

Economics is divided into two branches: microeconomics and macroeconomics. Microeconomics is the study of how individual households and firms make decisions and how they interact with one another in markets. Macroeconomics is the study of the economy as a whole.

There are a number of areas of controversy in macroeconomics. Some of the disagreements of economists stem from economic history and different schools of thought, which have shaped thinking over the last 250 years. It must be remembered that some of the most revered economic thinkers were people of their day, describing, analyzing and thinking about problems which were highly relevant at particular points in history. Economies and the context in which they operate do not stand still, and what was relevant and prescient about the way an economic system was thought to have worked in one period of history does not necessarily mean that its mechanics and analysis can be transferred to other time periods. For example, the analysis of the causes of the Great Depression of the 1930s and the policy options which were appropriate to remedy some of the problems caused by the crisis, may have some similarities to the causes and consequences of the Financial Crisis of 2007–9. However, the world in 2007 was a very different place from that of 1929–30. Given the differences, to what extent are policy measures which derived from the analysis of the Great Depression relevant and applicable to post-Crisis economies?

Before we look at measures of economic well-being, we are going to briefly introduce some key schools of thought which have applied analysis to the way the economy works. It is sensible to keep these in mind as we progress through the second half of the book and our study of macroeconomics.

MARXIST ECONOMICS

Karl Marx's analysis of economics and economic systems is based on the idea that 'modern' human history is dominated by class struggles and the exploitation of one class by another. This exploitation is based on economic power, slaves and slave owners, feudal lords and peasants, and ultimately, under a capitalist economic system, the bourgeoisie, capital-owning class and the proletariat. Marx had an explanation for the tendency of capitalism to experience repeated crises.

At the heart of his argument is the labour theory of value. Marx suggested that over a business cycle, the rate of profit would decline. He saw equilibrium not in terms of market clearing but the tendency for rates of profit on capital to equalize to capitalists who are in constant competition with each other. This market competition prevents capitalists from taking all surplus value for themselves; some of it must be reinvested to maintain the advantage over competitors. Some of this investment is in machinery, which Marx referred to as 'constant capital'.

Improvements in technology mean that this investment increases the ratio of constant capital to variable capital (labour). If this ratio increases at a faster rate than the ratio of surplus value to variable capital, there will be a tendency for the rate of profit to fall. Ultimately, this drive by capitalists to keep ahead of each other would drive some businesses out of the market, and unemployment (the reserve army of the unemployed) would increase as a result. Pressure on wages will cause wage rates to fall, causing the rate of profit to fall, and the process would begin over.

Ultimately, these crises would lead to workers taking to organizing themselves and ultimately taking the means of production into their own hands, possibly through revolution. One of the features of socialism, the social ownership and greater democratic control of the means of production, was a greater focus on equity as opposed to efficiency. This was a recognition that the outcomes of classical economic theory might be 'efficient' but not necessarily 'fair'. Marx was writing at a time when the amount of labour working in factories was extensive; it is perhaps no surprise that their latent power was to be feared. It is easy to see why the owning class was wary of the development of trade unions and organized labour. Marx's critique challenged the neo-classical orthodoxy and represented a challenge to the fundamental basis of the 'organized, civilized' society of the late nineteenth century.

THE AUSTRIAN SCHOOL

We have already encountered some of the economists associated with the Austrian School – Menger, Jevons and Walras. Menger wrote *Principles of Economics*, published in 1871, and argued that the analysis of economics, such as individual preferences and decisions at the margin, were universally applicable, much along the same lines that the classical economists had argued for using the 'law of demand and supply'. Menger's view was not accepted by other economists in Germany at the time, notably William Roscher, a professor at the University of Leipzig from 1848 until his death in 1894. It was, allegedly, some of Roscher's students who used the term 'Austrian school', as a means of insulting Menger and others on the faculty at the University of Vienna. Despite the insult, economists of the Austrian school have made significant contributions to economic thinking, although since the 1930s the Austrian school has been associated with universities other than Vienna, for example Chicago. In particular, Ludwig von Mises, Friedrich Hayek (the Nobel Prize winner for Economics in 1974 who taught at the University of Chicago) and Friedrich von Wieser have been at the forefront of an intellectual tradition that is linked to the Austrian school.

The fundamental beliefs associated with the Austrian school are those of the power of free markets, private property, property rights and individual choice, but tempered by an understanding of decision-making over time and under conditions of uncertainty. Austrian school economists recognize that ultimately individuals are the only ones that make choices, but in so doing generate unintended consequences. In analyzing choices, a focus on the process and relationships of exchange and bargaining is important. This reflects the idea that understanding human sciences like economics is not a case of being able to predict, but rather to understand. Recognition of the role of subjective views in economics is thus important in analyzing issues such as utility and opportunity cost.

Thinkers in the Austrian school tradition have made numerous contributions to macroeconomics, most notably in the field of monetary theory for which Hayek received his Nobel Prize. Austrian economists are noted for proposing free banking and the denationalization of money. Free banking would see the abolition of central banks, and a system where banks would compete in a free market, setting interest rates and issuing their own currency.

KEYNESIANISM

Keynesian economics is associated with the work of John Maynard Keynes (1883–1946). The very fact that a school of thought is named after him is perhaps a testament to the importance of Keynes. Keynes studied under Alfred Marshall and Arthur Pigou at Cambridge in the early 1900s, where he eventually ended up teaching. His most notable work, *The General Theory of Employment, Interest and Money*, published in 1936, has influenced subsequent generations of economists and presented a challenge to neo-classical and Austrian school assumptions that free markets, left to their own devices, clear. Keynes suggested that there was no reason that, for example, labour markets should clear, and that the persistence of unemployment over a period could be due to demand deficiencies in the economy. Demand deficiencies could be corrected by government intervention in the economy, by using tax and spending instruments to boost aggregate demand (or to reduce it when the economy was overheating). Keynes' ideas were developed by other notable economists including Joan Robinson, John Hicks, James Tobin, Arthur Okun, Robert Solow and Paul Samuelson.

Some of these developments were christened 'New Keynesianism', in particular, the work of Samuelson and Solow in developing the so-called 'neo-classical synthesis', which sought to explain the short-term failure of markets to clear because of sticky wages and prices. The attempt to explain Keynesian ideas in the context of neo-classical principles was coined 'bastard Keynesianism' by Joan Robinson.

A number of developed economies adopted Keynesian demand management policies after the Second World War, and unemployment remained relatively low and growth stable up until the late 1960s, when global pressures began to have wider economic impacts. The collapse of the Bretton Woods system of fixed exchange rates against gold, and the oil price shocks caused by the tensions between Arabs and Israelis in the early 1970s, began to reveal fault lines in economies.

Some argued that demand management policies had led to complacency among firms and workers, who expected governments to always come to the rescue when economic times got hard. This, it is argued, led to a lack of competitiveness and an over-reliance on the state. In the 1960s, research by economists into the role of money and monetary policy, and the problems being experienced by a number of countries of accelerating inflation and rising unemployment (stagflation), led to the decline in the use of demand management policies. Keynesian ideas became eclipsed by monetarism. The Financial Crisis 2007–9 and the subsequent global recession led to a resurgence of interest in Keynes' ideas.

MONETARISM

Monetarism has become associated with a number of things, not least an assumption that all monetarists are advocates of free markets. This is not the case. Monetarism is essentially linked to a number of beliefs, most notably that in the long run increases in the money supply have no effect on real variables (so-called monetary neutrality), but in the short run this neutrality may not exist. Monetary neutrality implies that an increase in the money supply will, over a period, lead to an increase in the price level, but not in real variables such as output, consumption and relative prices. The idea that changes in the money supply would lead to changes in the price level was encapsulated in the Fisher equation of exchange.

One of the leading economists of the monetarist school was Milton Friedman, winner of the Nobel Prize for Economics in 1976. Friedman had challenged aspects of Keynes' ideas when he published *A Theory of the Consumption Function* in 1957, which developed the idea of the permanent income hypothesis. In other work, he researched price determination in individual markets, but his association with the revival of monetarism is perhaps what he will be remembered for. Friedman noted that 'inflation is always and everywhere a monetary phenomenon'. The subjugation of Keynesian demand management policies in the 1970s as inflation accelerated alarmingly in a number of developed economies, and the acceptance of Friedman's analysis of the role of the money supply, led to a greater focus on attempting to control the money supply as a means of controlling inflation.

In the UK and the United States, the respective governments announced targets for the growth in the money supply throughout much of the 1980s, but in the event failed to achieve these targets. The policy was abandoned in favour of exchange rate targeting and, later, inflation targeting.

THE NATURE OF MACROECONOMICS

The goal of macroeconomics is to explain the economic changes that affect many households, firms and markets simultaneously. Economists analyze the economy as a whole, as it provides an indication of the overall well-being. We can identify two measures of well-being. Subjective well-being refers to the way in which people evaluate their own happiness, and objective well-being refers to measures of the quality of life and uses indicators such as educational attainment, measures of the standard of living, life expectancy and so on. Many of the macroeconomic variables which economists study are inextricably linked with these definitions of well-being. Having a reasonable income means that people can afford to buy the necessities and luxuries in life which contribute to subjective well-being. Countries with higher national incomes can provide better quality education, more secure jobs, higher quality of housing, infrastructure and healthcare, among other things, which affect objective well-being.

One of the most common ways in which the well-being of an economy is measured is through *gross domestic product*, or simply GDP, which measures the total income of a nation. GDP is a widely reported macroeconomic statistic, but it has its critics, and the way the data are collected also has limitations. As with many things, what is considered a good measure of well-being depends on underlying belief systems and judgements. For example, GDP measures focus on incomes, with the implication that higher national income equates to an increased ability to acquire more goods and services, which in turn means increased well-being. This can be interpreted as being based on a consumerist value system, which may not take adequate account of factors which can also contribute to subjective and objective well-being. Some economists have pointed out that despite considerable increases in GDP since the late 1960s in many developed countries, reports of subjective well-being have not risen by the same amount. There may be a number of reasons for this, but it seems that what economists and politicians might use as a measure of the health and wealth of an economy are not perceived in the same way by the citizens of a country.

We will begin our investigation of measures of well-being by looking at national income. Given the caveats outlined above, this remains one of the most important ways in which well-being is reported and measured.

THE ECONOMY'S INCOME AND EXPENDITURE

Incomes are used as a measure of well-being because incomes can be used to purchase life's necessities and luxuries. Income is, therefore, equated with the standard of living. Recall that the standard of living is defined as the amount of goods and services that can be purchased by the population of a country. By this definition, those with higher incomes enjoy higher standards of living – better housing, better healthcare, fancier cars, more frequent and luxurious holidays and so on. The same logic can be applied to a nation's overall economic well-being. To judge whether an economy is doing well or poorly, we can look at the total income that everyone in the economy is earning. This is the task of GDP.

GDP measures two things at once: the total income of everyone in the economy and the total expenditure on the economy's output of goods and services. These two things are really the same because every transaction has two parties: a buyer and a seller. Every euro of spending by some buyer is a euro of income for a seller. For example, if Michael pays Astrid €20 to clean the windows at his house, Astrid is a seller of a service and Michael is a buyer. Astrid earns an income of €20 and Michael spends €20. Thus the transaction contributes equally to the economy's income and to its expenditure. GDP, whether measured as total income or total expenditure, rises by €20.

The Circular Flow of Income

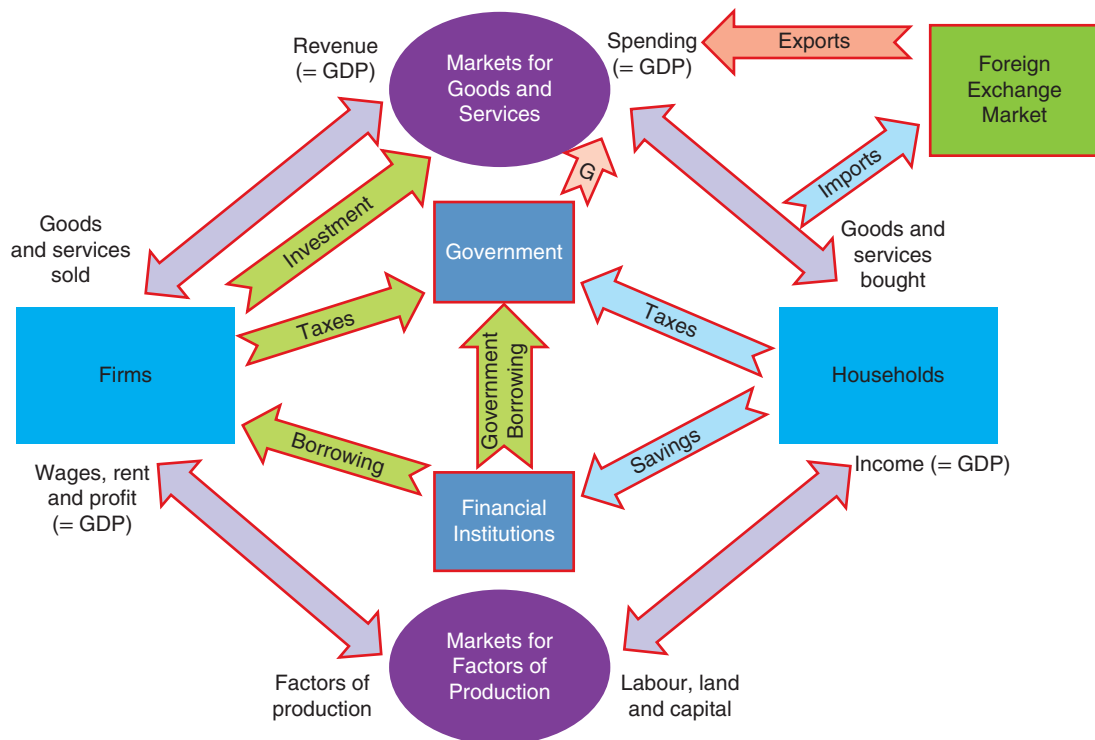
The equality of income and expenditure can be seen through the circular flow diagram in Figure 20.1. This is a model which describes all the transactions between households and firms in a simple economy. In this economy, the market for goods and services is where households and firms interact through firms providing goods and services which are bought by households. The spending by households on these goods and services represents the revenue of firms. The two also interact in the market for factors of production. Firms use the money they receive from the sale of goods and services to purchase the factors of production (land, labour, capital and enterprise) from households who provide these factor services. The factor incomes received by households (wages, rent, interest and profit) are used to pay for goods and services. In this economy, money flows from households to firms and then back to households.

Not all of the incomes households receive are spent in the market for goods and services. Some of this income is subject to tax which is paid to the government. Some income will be saved, and the funds find their way into financial institutions in the form of pension saving, insurance and assurance, and saving in bank accounts. Some of the income spent on goods and services by households leaves the economy in the form of spending on imports. Similarly, some of the income (Y) received by firms is paid to the government in business taxes. Taxes (T), saving (S) and spending on imports (M) are referred to as *leakages* from the circular flow.

FIGURE 20.1

The Circular Flow Diagram

Households buy goods and services from firms, and firms use the revenue from sales to pay wages to workers, rent to landowners, interest on capital and profit to owners of firms. GDP equals the total amount spent by households in the market for goods and services. It also equals the total wages, rent, interest and profit paid by firms in the markets for the factors of production. Leakages to the circular flow include taxes, savings and spending on imports whereas injections to the circular flow include government spending, investment and revenue generated from exports.



Governments use tax revenue and borrowing from financial institutions to spend on government services and investment on infrastructure, education, health, defence and so on, representing government spending (G) which goes back into the circular flow. Firms make use of the funds provided by savers in financial institutions through borrowing to finance investment spending on new plant, equipment and expansion. This investment (I) finds its way back into the market for goods and services. Firms will also sell some of the goods and services they produce abroad, and so the revenue from exports (X) flows back into the system. Financial institutions also lend money overseas and firms overseas invest money into the country which is recorded as net capital outflow (NCO). Government spending (G), investment spending (I) and revenue from exports (X) are referred to as *injections* to the circular flow.

In theory, measuring GDP through income or expenditure will arrive at the same result. National income (NY), national expenditure and GDP are invariably used synonymously. A common measure of showing GDP is to add up the sum of expenditures in the economy – consumer spending (C), investment spending (I), government spending (G) and the difference between the amount paid out on imports (M) and that received in exports (X) (net exports: $X - M$ or NX). This provides a measure of GDP given by the following equation:

$$GDP(NY) = C + I + G + NX$$

SELF TEST What two things are measured by GDP? How can it measure two things at once?

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

Gross domestic product (GDP) is the market value of all final goods and services produced within a country in a given period of time.

gross domestic product (GDP) the market value of all final goods and services produced within a country in a given period of time

Let's consider each phrase in this definition.

'GDP Is the Market Value ...'

GDP adds together many different kinds of products into a single measure of the value of economic activity. To do this, it uses market prices. Because market prices measure the amount people are willing to pay for different goods, they reflect the value of those goods. If the price of an apple is twice the price of an orange, then an apple contributes twice as much to GDP as does an orange.

'... Of All ...'

GDP includes all items produced in the economy and sold in markets. GDP measures the market value of not just apples and oranges but also pears and grapefruit, books and films, haircuts, healthcare and so on.

GDP also includes the market value of the housing services provided by the economy's stock of housing. For rental housing, this value is easy to calculate – the rent equals both the tenant's expenditure and the landlord's income. Yet many people own the place where they live and, therefore, do not pay rent. The government includes this owner occupied housing in GDP by estimating its rental value. That is, GDP is based on the assumption that the owner, in effect, pays rent to themselves, so the rent is included both in their expenditure and income.

There are many goods and services that are traded, but not through official recorded markets, and these transactions may not appear in GDP figures. For example, vegetables bought at the greengrocer's shop or the supermarket are part of GDP, but vegetables grown in your garden are not. There are many trades which are carried out on a cash basis and hidden from the authorities. For example, a plumber might do a repair on a boiler, be paid in cash and not declare this as part of their income, or a teenager child-minds for a couple on a Saturday night and gets paid, but the value of this work is hidden from GDP figures. This type of activity is referred to as the 'informal', 'shadow' or 'black' economy. It is extremely difficult to measure the size of the shadow economy, but some studies put the number of people engaging in shadow work at 30 million in the EU, and anything up to a quarter of a country's GDP. Some estimates have put the size of the shadow economy in the UK at around £150bn (£190bn) a year and across Europe as a whole, around €2 trillion.

There are also transactions of goods and services in markets which are illegal and, as such, are unlikely to be declared to tax authorities. This can also affect the value of GDP. From September 2014, the UK's national statistical service, the Office for National Statistics (ONS), included estimates of the level of activity in illegal drugs and prostitution to bring the figures for the UK National Accounts in line with methods used across other EU countries. The ONS estimated that the average impact on the UK economy by including illegal drugs and prostitution amounts to around £11 billion (£12.3 billion) per year.

A number of economists have noted that calculations of GDP do not take into account the value of work carried out by housewives and househusbands, and neither is the value of childcare work carried out by grandparents, for example. Feminist economists, in particular, criticize the distinction between 'productive' work carried out in markets and 'unproductive work' which is not. The work done in supporting families in the home can contribute significantly to overall well-being and is the centre of most people's lives. This contribution is not considered in GDP calculations, and as a result not only undervalues the true measure of well-being in the economy but also the value of women's productive contribution to the economy.

'... Final ...'

When a paper company sells paper to a greetings card company, the paper is called an *intermediate good*, and the card is called a *final good*. GDP includes only the value of final goods. The reason is that the value of intermediate goods is already included in the prices of final goods. Adding the market value of the paper to the market value of the card would be double counting. That is, it would (incorrectly) count the paper twice.

An important exception to this principle arises when an intermediate good is produced and, rather than being used, is added to a firm's inventory of goods to be used or sold at a later date. In this case, the intermediate good is taken to be 'final' for the moment, and its value as inventory investment is added to GDP. When the inventory of the intermediate good is later used or sold, the firm's inventory investment is negative, and GDP for the later period is reduced accordingly.

'... Goods and Services ...'

GDP includes both tangible goods (food, clothing, cars) and intangible services (haircuts, house cleaning, doctor's visits). When you (legally) download an album by your favourite band, you are buying a good, and the purchase price is part of GDP. When you pay to hear a concert by the same band, you are buying a service, and the ticket price is also part of GDP.

'... Produced ...'

GDP includes goods and services currently produced. It does not include transactions involving items produced in the past. When you buy a new textbook for your economics course, the value of the textbook is included in the GDP of the country in which the publisher operates. When one person sells a used textbook to another person, the value of the used textbook is not included in GDP.

‘... Within a Country ...’

GDP measures the value of production within the geographic confines of a country. When an Australian citizen works temporarily in the UK, their production is part of UK GDP. When a UK citizen owns a factory in Bulgaria, the production at their factory is not part of UK GDP (it's part of Bulgaria's GDP). Thus items are included in a nation's GDP if they are produced domestically, regardless of the nationality of the producer.

‘... In a Given Period of Time’

GDP measures the value of production that takes place within a specific interval of time. Usually that interval is a year or a quarter of a year (three months). GDP measures the economy's flow of income and expenditure during that interval.

When the government reports the GDP for a quarter, it usually presents GDP ‘at an annual rate’. This means that the figure reported for quarterly GDP is the amount of income and expenditure during the quarter multiplied by four. The government uses this convention so that quarterly and annual figures on GDP can be compared more easily.

In addition, when the government reports quarterly GDP, it presents the data after they have been modified by a statistical procedure called *seasonal adjustment*. The unadjusted data show clearly that the economy produces more goods and services during certain times of the year than during others (as you might guess, December's holiday shopping season is a high point in many countries, while the period before and after Ramadan is a high point for many Muslim countries). When monitoring the condition of the economy, economists and policymakers often want to look beyond these regular seasonal changes. Therefore government statisticians adjust the quarterly data to take out the seasonal cycle.

SELF TEST Which contributes more to GDP – the production of £1 of tin or the production of £1 of gold? Why?

FYI



Other Measures of Income

When the ONS or Eurostat computes the GDP every three months for the UK and the EU respectively, they also compute various other measures of income to arrive at a more complete picture of what's happening in the economy. These other measures differ from GDP by excluding or including certain categories of income. What follows is a brief description of five of these income measures, ordered from largest to smallest, based on the UK.

- **Gross national product (GNP)** is the total income earned by a nation's permanent residents (called nationals). It differs from GDP by including income that domestic citizens earn abroad and excluding income that foreigners earn in the domestic country, as we saw above. For most countries, domestic residents are responsible for most domestic production, so GDP and GNP are quite close.
- **Net national product (NNP)** is the total income of a nation's residents (GNP) minus losses from depreciation. *Depreciation* is the wear and tear on the economy's stock of equipment and structures, such as lorries rusting and computers becoming obsolete.
- **National income (NY)** is the total income earned by a nation's residents in the production of goods and services. It differs from net national product by excluding indirect business taxes (such as sales taxes) and including business subsidies. NNP and NY also differ because of a 'statistical discrepancy' that arises from problems in data collection.

(Continued)

- *Personal income* is the income that households and non-corporate businesses receive. Unlike *NY*, it excludes *retained earnings*, which is income that corporations have earned but have not paid out to their owners. It also subtracts corporate income taxes and contributions for social insurance. In addition, personal income includes the interest income that households receive from their holdings of government debt and the income that households receive from government transfer programmes, such as welfare and social security payments.
- *Disposable personal income* is the income that households and non-corporate businesses have left after satisfying all their obligations to the government. It equals personal income minus personal taxes and certain non-tax payments (such as parking tickets).

Although the various measures of income differ in detail, they almost always tell the same story about economic conditions. When GDP is growing rapidly, these other measures of income are usually growing rapidly; when GDP is falling, these other measures are usually falling as well.

THE COMPONENTS OF GDP

Spending in the economy takes many forms. At any moment, the Müller family may be having lunch in a Munich restaurant; Honda may be building a car factory on the banks of the Rhine; and the German army may be procuring weapons from German arms manufacturers. German GDP includes all of these various forms of spending on domestically produced goods and services. Similarly, each country in Europe will monitor the forms of spending and income to arrive at the GDP for that country.

Earlier we noted that national income could be described by the equation:

$$Y \equiv C + I + G + NX$$

Notice that we have used the three bar, 'identically equals' symbol, ' $\equiv$ ', in this equation. This is because the equation is an *identity* – an equation that must be true by the way the variables in the equation are defined. In this case, because each euro or pound of expenditure included in GDP is placed into one of the four components of GDP, the total of the four components must be equal to GDP. For the most part we'll follow normal practice in dealing with identities and use the usual equals sign, ' $=$ '. Let's look at each of these four components more closely.

Consumption

Consumption is spending by households on goods and services. 'Goods' include household spending on durables, such as cars and appliances like washing machines and fridges, and non-durable goods, such as food and clothing.

consumption spending by households on goods and services, with the exception of purchases of new housing

'Services' include such intangible items as haircuts, entertainment and medical care. Household spending on education is also included in consumption of services (although one might argue that it would fit better in the next component).

Investment

Investment is the purchase of buildings, equipment and machinery that will contribute to future productive output. It is the sum of purchases of capital equipment, inventories and structures. Investment in structures includes expenditure on new housing. By convention, the purchase of a new house is the one form of household spending categorized as investment rather than consumption.

investment spending on capital equipment, inventories and structures, including household purchases of new housing

The treatment of inventory accumulation is noteworthy. When Sony produces a smartphone and, instead of selling it, adds it to its inventory, Sony is assumed to have ‘purchased’ the phone for itself. That is, the national income accountants treat the phone as part of Sony’s investment spending. If Sony later sells the phone out of inventory, Sony’s inventory investment will then be negative, offsetting the positive expenditure of the buyer. Inventories are treated this way because one aim of GDP is to measure the value of an economy’s production, and goods added to inventory are part of that period’s production.

Government Spending

Government spending includes spending on goods and services by local and national governments. It includes the salaries of government workers and spending on public works.

government spending spending on goods and services by local and national governments

The meaning of ‘government spending’ requires a little clarification. When the government pays the salary of an army general, that salary is part of government spending. What happens when the government pays a social security benefit to one of the elderly? Such government spending is called a **transfer payment** because it is not made in exchange for a currently produced good or service. Transfer payments alter household income, but they do not reflect the economy’s production (from a macroeconomic standpoint, transfer payments are like negative taxes). Because GDP is intended to measure income from, and expenditure on, the production of goods and services, transfer payments are not counted as part of government spending.

transfer payment a payment for which no good or service is exchanged

Net Exports

Net exports equal the purchases of domestically produced goods and services by foreigners (exports), which generates a flow of funds into a country, minus the domestic purchases of foreign goods (imports), which result in funds leaving a country. A domestic firm’s sale to a buyer in another country, such as the sale of Sony phones to customers in Europe, increases Japanese net exports.

net exports spending on domestically produced goods and services by foreigners (exports) minus spending on foreign goods by domestic residents (imports)

The ‘net’ in ‘net exports’ refers to the fact that the value of imports is subtracted from the value of exports. This subtraction is made because imports of goods and services are included in other components of GDP. For example, suppose that a UK household buys a £30,000 car from Volvo, the Swedish car maker. That transaction increases consumption in the UK by £30,000 because car purchases are part of consumer spending in the UK. It also reduces net exports by £30,000 because the car is an import (note it represents an export for Sweden). In other words, net exports include goods and services produced

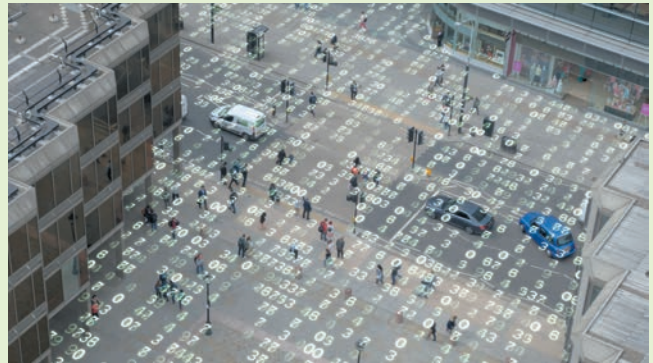
abroad (with a minus sign) because these goods and services are included in consumption, investment and government purchases (with a plus sign). Thus when a domestic household, firm or government buys a good or service from abroad, the purchase reduces net exports – but because it also raises consumption, investment or government purchases, it does not affect GDP. The above example shows the importance of making sure that we focus on a particular country when discussing imports and exports because of the potential for confusion to arise.

Once GDP figures are published, they can be presented in different ways. One of the most common is GDP per head or per capita. GDP per capita is found by dividing the GDP of a country by the population of that country to express national income per head of the population. This measure is useful in comparing GDP across different countries.

CASE STUDY Revisions to GDP Figures

National or regional statistical services such as the ONS or Eurostat have very sophisticated systems for collecting and calculating national income data. The European system of national and regional accounts (known as ESA95) compiles data on the structure and developments of member state economies. It uses an agreed accounting framework which is recognized across the world to allow each economy to be described and to be able to draw accurate and reliable comparisons between different regions and economies or groups of economies. It is worth trying to imagine how challenging a task collecting all this data actually is. Just look around you whenever you are out at the amount of economic activity which is taking place – people buying and selling goods and services in shops, there are online transactions, people travelling on public transport, people using their cars for which they have bought insurance, fuel and paid for maintenance, road sweepers cleaning up litter, emptying bins and removing chewing gum from the pavements, police patrolling the streets and so on. Trying to collect all the data for the expenditures that are taking place is a massive task. Inevitably, any published GDP figure is provisional and statistical services continue to update the data they receive and publish revised figures on a regular basis.

Providing accurate data is important, because business, government and investment decisions are all made with the macroeconomic environment in mind. It is made very clear, therefore, that GDP data when published is subject to revision and these revisions can sometimes be significant. Initial figures for quarterly GDP are referred to as 'flash estimates' because they are based on a fraction of the total data used for final estimate. The ONS, for example, will provide at least two revisions to initial estimates. Once all the data are in, initial estimates can be confirmed or revised up or down. Announcements on GDP data, therefore, need to be looked at with some caution – one quarter's figures may not necessarily be a reliable indicator of the pattern of GDP over time.



Data flows in from the economy all the time and results in the necessity to revise GDP figures.

SELF TEST List the four components of expenditure. Which of these do you think accounts for the largest proportion of GDP in a country? Why?

REAL VERSUS NOMINAL GDP

Changes in total expenditure on goods and services from one year to another can be looked at in two ways:

1. The economy is producing a larger output of goods and services (a real increase); or
2. Goods and services are being sold at higher prices (a nominal increase).

When studying changes in the economy over time, these two effects are separated. The focus is on a measure of the total quantity of goods and services the economy is producing that is not affected by changes in the prices of those goods and services.

To do this, economists use a measure called **real GDP**. Real GDP answers a hypothetical question: what would be the value of the goods and services produced this year if we valued these goods and services at the prices that prevailed in some specific year in the past? By evaluating current production using prices that are fixed at past levels, real GDP shows how an economy's overall production of goods and services changes over time. The GDP figures produced using this method are called **GDP at constant prices**. GDP figures produced without taking into consideration the change in prices over time are called **GDP at current or market prices** and are calculated by taking the output of goods and services and multiplying by the price of those goods and services in the reporting year.

real GDP the production of goods and services valued at constant prices

GDP at constant prices gross domestic product calculated using prices that existed at a particular base year, which takes into account changes in inflation

GDP at current or market prices gross domestic product calculated by multiplying the output of goods and services by the price of those goods and services in the reporting year

To see more precisely how real GDP is constructed, let's consider an example.

A Numerical Example

Table 20.1 shows some data for an economy that produces only two goods – apples and potatoes. The table shows the quantities of the two goods produced and their prices in the years 2019, 2020 and 2021.

TABLE 20.1 Real and Nominal GDP

Prices and quantities				
Year	Price of apples per kg (€)	Quantity of apples (kg)	Price of potatoes per kg (€)	Quantity of potatoes (kg)
2019	1	100	2	50
2020	2	150	3	100
2021	3	200	4	150
Year	Calculating nominal GDP			
2019	$(\text{€1 per kg apples} \times 100 \text{ kg}) + (\text{€2 per kg potatoes} \times 50 \text{ kg}) = \text{€200}$			
2020	$(\text{€2 per kg apples} \times 150 \text{ kg}) + (\text{€3 per kg potatoes} \times 100 \text{ kg}) = \text{€600}$			
2021	$(\text{€3 per kg apples} \times 200 \text{ kg}) + (\text{€4 per kg potatoes} \times 150 \text{ kg}) = \text{€1,200}$			
Year	Calculating real GDP (base year 2019)			
2019	$(\text{€1 per kg apples} \times 100 \text{ kg}) + (\text{€2 per kg potatoes} \times 50 \text{ kg}) = \text{€200}$			
2020	$(\text{€1 per kg apples} \times 150 \text{ kg}) + (\text{€2 per kg potatoes} \times 100 \text{ kg}) = \text{€350}$			
2021	$(\text{€1 per kg apples} \times 200 \text{ kg}) + (\text{€2 per kg potatoes} \times 150 \text{ kg}) = \text{€500}$			
Year	Calculating the GDP deflator			
2019	$(\text{€200}/\text{€200}) \times 100 = 100$			
2020	$(\text{€600}/\text{€350}) \times 100 = 171$			
2021	$(\text{€1,200}/\text{€500}) \times 100 = 240$			

To compute total spending in this economy, we would multiply the quantities of apples and potatoes by their prices. In the year 2019, 100 kg of apples are sold at a price of €1 per kg, so expenditure on apples equals €100. In the same year, 50 kg of potatoes are sold for €2 per kg, so expenditure on potatoes also equals €100. Total expenditure in the economy – the sum of expenditure on apples and expenditure on potatoes – is €200. This amount, the production of goods and services valued at current or market prices (i.e. the price existing in the reporting year), is called **nominal GDP**.

nominal GDP nominal GDP the production of goods and services valued at current prices

The table shows the calculation of nominal GDP for these three years. Total spending rises from €200 in 2019 to €600 in 2020 and then to €1,200 in 2021. Part of this rise is attributable to the increase in the quantities of apples and potatoes produced, and part is attributable to the increase in the prices of apples and potatoes.

To obtain a measure of the amount produced that is not affected by changes in prices, we use real GDP, the production of goods and services valued at constant prices. Real GDP is calculated by first choosing one year as a *base year*. We then use the prices of apples and potatoes in the base year to compute the value of goods and services in all of the years. In other words, the prices in the base year provide the basis for comparing quantities in different years.

Suppose that we choose 2019 to be the base year in our example. We can then use the prices of apples and potatoes in 2019 to compute the value of goods and services produced in 2019, 2020 and 2021. Table 20.1 shows these calculations. To compute real GDP for 2019 we use the prices of apples and potatoes in 2019 (the base year) and the quantities of apples and potatoes produced in 2019. (For the base year, real GDP always equals nominal GDP.) To compute real GDP for 2020, we use the prices of apples and potatoes in 2019 (the base year) and the quantities of apples and potatoes produced in 2020. Similarly, to compute real GDP for 2021, we use the prices in 2019 and the quantities in 2021. When we find that real GDP has risen from €200 in 2019 to €350 in 2020 and then to €500 in 2021, we know that the increase is attributable to an increase in the *quantities produced*, because the prices are being held fixed at base year levels.

The growth rate in real GDP takes the difference between GDP across the two time periods under consideration denoted by $GDP_t - GDP_{t-1}$, divided by GDP in year $t-1$. Multiplying the result by 100 gives the percentage rate of growth in real GDP.

$$\text{Growth rate of real GDP in year } t = \frac{(GDP_t - GDP_{t-1})}{GDP_{t-1}} \times 100$$

The GDP Deflator

Nominal GDP reflects both the prices of goods and services and the quantities of goods and services the economy is producing. In contrast, by holding prices constant at base year levels, real GDP reflects only the quantities produced. From these two statistics we can compute a third, called the **GDP deflator**, sometimes also referred to as the *implicit price level*, which reflects the prices of goods and services but not the quantities produced.

GDP deflator a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100

The GDP deflator is calculated as follows:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Because nominal GDP and real GDP must be the same in the base year, the GDP deflator for the base year always equals 100. The GDP deflator for subsequent years measures the change in nominal GDP from

the base year that cannot be attributable to a change in real GDP by measuring the current level of prices relative to the level of prices in the base year.

Imagine that the quantities produced in the economy rise over time, but prices remain the same. In this case, both nominal and real GDP rise together, so the GDP deflator is constant. Now suppose that prices rise over time, but the quantities produced stay the same. In this second case, nominal GDP rises but real GDP remains the same, so the GDP deflator rises as well. Notice that, in both cases, the GDP deflator reflects what's happening to prices, not quantities. Real GDP is found by taking the nominal GDP and dividing by the GDP deflator.

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP Deflator}}$$

Let's now return to our numerical example in Table 20.1. The GDP deflator is computed at the bottom of the table. For 2019, the base year, nominal GDP is €200, and real GDP is €200, so the GDP deflator is 100. For the year 2020, nominal GDP is €600, and real GDP is €350, so the GDP deflator is 171. Because the GDP deflator rose in year 2020 from 100 to 171, we can say that the price level increased by 71 per cent.

The GDP deflator is one measure that economists use to monitor the average level of prices in the economy. We examine another – the consumer prices index – later in this chapter, where we also describe the differences between the two measures.

SELF TEST Define real and nominal GDP. Which is a better measure of economic well-being? Why?

Chain Linking

We have looked at measuring GDP through income and expenditure methods and said that in theory they are the same. GDP can also be measured by looking at the output or production method. The price of a product represents the value of the inputs that went into production. At each stage of the production process the value of output can be recorded – the value added. The ONS defines **gross value added** (GVA) as the 'contribution to the economy of each individual producer, industry or sector'. GVA is used as the measure of the output and income approaches to measuring GDP. Statistics offices usually classify output according to industry types – construction, agriculture forestry and fisheries, mining, services and so on.

gross value added the contribution of domestic producers, industries and sectors to an economy

The measure for GDP is a 'volume measure', measuring 'how much' has been produced. In the UK the ONS must produce a single measure of GDP and does this by using the three approaches – income, expenditure and output – with the final figure reflecting the three measures. In theory these measures should all be the same, but the process of collecting information from millions of transactions means that in reality they would never be exact.

Imagine a situation, for example, where someone has some work done on their house – say the plastering of a room. The total price for the work paid by the owner of the house might be £1,500 – this should represent the sum of the resources used in the job. The plasterer may declare to Her Majesty's Revenue and Customs (HMRC, that is responsible for collecting UK taxes) that the work represented income of £1,200 (to evade paying income tax – illegal, but nevertheless it happens), and the resources used might have included the plasterboard, plaster, electricity, buckets, nails, water, trowels and hawks, the vehicles to get to and from the builders' merchants, etc. It is easy to see how, even in such a simple example, it becomes incredibly difficult to keep track of every item involved!

The choice of which base year to use in calculating real GDP is also important. Up until early 2004, the ONS used a method referred to as 'fixed base aggregations'. For example, it might have compared the data using 1995 as the base year. However, as with any statistical data which use a base year in their compilation, this can lead to inconsistencies because circumstances change. Goods, for example, that were common in 1995 may not exist anymore and there might be some products that were not available at that

time – Blu-ray DVDs, for example. In general, the more you can update the base year, the more accurate the statistics will be.

Fixed base aggregations in GDP statistics were used as follows. The growth of the different parts of the economy was given weights when trying to arrive at the final figure for GDP. This takes account of their relative importance in the economy and a base year of 1995 was used to help determine these weights. These base years were updated every five years.

The method, known as '**annual chain linking**', helps to overcome these problems. Rather than updating the base year every five years, this method does it every year, calculating the prices in previous years' prices (PYPs). You will see GDP tables expressed as 'GDP (CVM)', where CVM means Chained Volume Measures, as opposed to 'GDP in constant prices'. The ONS and other statistical services are constantly looking to update their methods to provide more reliable and accurate data, and this is one example of such a revision of methods.

annual chain linking a method of calculating GDP volume measures based on prices in the previous year

Table 20.2 shows an example of how the chain linking method works. We will use our previous example of an economy producing apples and potatoes in the years 2019 and 2020 with the following prices and quantities.

TABLE 20.2 Chain Linking

Year	Price of apples per kg (€)	Quantity of apples per kg	Price of potatoes per kg (€)	Quantity of potatoes
2019	1	100	2	50
2020	2	150	3	100

$$2019 \text{ nominal GDP} = (100 \times €1) + (50 \times €2) = €200$$

$$2020 \text{ nominal GDP} = (150 \times €2) + (100 \times €3) = €600$$

If 2019 is used as the base year, the value of 2019 and 2020 production at 2019 prices is as follows:

$$2019: \text{Nominal GDP} = \text{Real GDP} = €200$$

$$2020: \text{Real GDP} = (150 \times €1) + (100 \times €2) = €350$$

$$\text{The percentage change in real GDP} = \left(\frac{(350 - 200)}{200} \right) \times 100 = 75\%$$

Chain-weighted real GDP uses average prices over the time period:

$$\text{Average price of apples between 2019 and 2020} = \frac{(€1 + €2)}{2} = €1.50$$

$$\text{Average price of potatoes between 2019 and 2020} = \frac{(€2 + €3)}{2} = €2.50$$

$$2019 \text{ GDP at average prices} = (100 \times €1.50) + (50 \times €2.50) = €275$$

$$2020 \text{ GDP at average prices} = (150 \times €1.50) + (100 \times €2.50) = €475$$

$$\text{Percentage change in chain-weighted GDP} = \left(\frac{(€475 - €275)}{275} \right) \times 100 = 72.7\%$$

$$\text{Implicit GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$\text{In 2019, the implicit GDP deflator} = \frac{200}{200} \times 100 = 100$$

$$\text{In 2020, the implicit GDP deflator} = \frac{600}{350} \times 100 = 171.4$$

$$\text{The percentage change in the implicit GDP deflator} = 71.4\%$$

With chain weighting, the base year is the mid-point between the two years.

$$\begin{aligned}\text{The 2019 chain-weighted GDP deflator} &= \frac{\text{Nominal GDP}_{2019}}{\text{Real GDP at 2019 average prices}} \times 100 \\ &= \left(\frac{200}{275} \right) \times 100 = 72.7\end{aligned}$$

$$\begin{aligned}\text{The 2020 chain-weighted GDP deflator} &= \frac{\text{Nominal GDP}_{2020}}{\text{Real GDP at 2020 average prices}} \times 100 \\ &= \left(\frac{600}{475} \right) \times 100 = 126.3\end{aligned}$$

$$\text{The percentage change in the chain-weighted GDP deflator} = \left(\frac{126.3 - 72.7}{72.7} \right) \times 100 = 73.7\%$$

In terms of analysis of the data it should make little difference, but recognizing it enables you to be more confident that the data are more likely to be accurate and reliable compared to the previous measures. The vast majority of official data is produced using chain-weighted measures.

THE LIMITATIONS OF GDP AS A MEASURE OF WELL-BEING

The use of GDP as a measure of well-being is well established. Yet some people dispute the validity of GDP as a measure of well-being. Critics of the use of GDP suggest that it is too focused on material possessions and income. They argue that there are many things that are not measured by GDP but contribute to the quality of life and economic well-being, such as the health of a country's children, or the quality of their education, or even the beauty of the poetry making up their literary heritage.

To counter this, it can be argued that GDP might not measure these things, but nations with larger GDPs can afford better healthcare, better educational systems and can afford to teach more of their citizens to read and enjoy poetry. GDP does not take account of our intelligence, integrity, courage or wisdom, but all these laudable attributes are easier to foster when people are less concerned about being able to afford the material necessities of life. In short, GDP does not directly measure those things that make life worthwhile, but it does measure our ability to obtain the inputs into a worthwhile life.

It is widely recognized, however, that GDP is not a perfect measure of well-being. Some things that contribute to a good life are left out of GDP.

- *Leisure.* Suppose, for instance, that everyone in the economy suddenly started working every day of the week, rather than enjoying leisure at the weekends. More goods and services would be produced, and GDP would rise. Yet, despite the increase in GDP, we should not conclude that well-being had improved. The loss from reduced leisure would offset the gain from producing and consuming a greater quantity of goods and services.
- *Work in the home and volunteer work.* Because GDP uses market prices to value goods and services, it excludes the value of much activity that takes place outside of markets. In particular, GDP omits the value of goods and services produced at home. Volunteer work also contributes to the well-being of those in society, but GDP does not reflect these contributions.
- *The quality of the environment.* Imagine that the government eliminated all environmental regulations. Firms could then produce goods and services without considering the pollution they create, and GDP might rise. Yet well-being would most likely fall. The deterioration in the quality of air and water would more than offset the gains from greater production.
- *Distribution of incomes.* A society in which 100 people have annual incomes of €50,000 has GDP of €5 million and, not surprisingly, GDP per capita of €50,000; so does a society in which 10 people earn €500,000 and 90 suffer with nothing at all. Few people would look at those two situations and call them equivalent. GDP per person tells us what happens to the average person, but behind the average lies a large variety of personal experiences. GDP does not tell us much about income distributions.

The Economics of Happiness

Despite the massive increase in wealth, incomes and access to material goods and services for many people in developed countries since the late 1960s, our perception of happiness has not really changed that much. Numerous surveys have highlighted relatively stable rates of 'happiness' in rich countries. Professor Richard Layard, one of a group of economists including Andrew Oswald, Stephen Nickell, Robert Skidelsky, Tim Besley, Will Hutton, and those behavioural economists that we have met before in this book, have studied this apparent paradox. Layard states that in relation to Western societies: 'on average, people are no happier than they were fifty years ago' (Layard, R. (2005) *Happiness: Lessons from a New Science*. London: Penguin).

Psychologists and economists have found that links between how people see their own happiness and other things that might influence their judgement are statistically strong. This also seems to apply across different countries. As a result, we might be able to conclude that there are a range of factors that can contribute to a definition of happiness; if you are fortunate enough to find yourself being able to boast of having these characteristics, or in some cases managing to avoid them, then you are more likely to be happy.

Layard identified some key factors that may contribute to 'happiness'. He suggested that some of the factors associated with promoting happiness include sex, socializing, relaxing, praying, worshipping or meditating, eating, exercising, watching TV and shopping, among others. Other studies have suggested that an individual's level of education, health, whether they are married, single or divorced, the level of income enjoyed, whether they are working/unemployed/retired, their aspirations, and whether they have experienced bereavement, can all be contributory factors.

From such characteristics, equations to statistically arrive at measures of well-being can be derived. These have been used by both economists and psychologists and have been found to be surprisingly reliable in statistical terms. One of the leading thinkers on the economics of happiness is Professor Andrew Oswald who is based at the University of Warwick. Oswald offers the following formula:

$$W_{it} = \alpha + \beta X_{it} + \varepsilon_{it}$$

In this formula, W_{it} is the reported well-being of an individual at a particular time period, X represents a collection of variables that are known to be characteristics affecting well-being at a particular time period: these could be economic, such as income, or demographic (such as gender). The final term is an error term, which is used to take into account unobserved factors that may have an impact on the final outcome.

One of the factors above that may be affecting happiness levels more than we might expect is aspirations. You are very likely, whatever your age, to have been regaled by your parents that things were different in their day and that 'people these days have so much more'. That is very true, but what might also be the case is that different generations are starting to expect more. If your parents have managed to reach middle age and have a comfortable Jaguar to drive around in, you might start to expect such a vehicle to be the norm and would hope it might be your first car and so on.

Layard suggests that unhappiness results from society viewing how it is developing in terms of a zero-sum game. What he means by this is that we might increasingly view the scramble to gain money and status in terms of a competitive game that has a winner and a loser. If one person gets a high ranking job with a large salary and lots of status, it means that another person somehow loses out – either by not being able to get that same job, or in some sort of psychological way. Such a perception of life being a zero-sum game is a source of much unhappiness. This reflects the idea of *positional externalities*.

If we know something of the factors that can contribute to making someone 'happy' and also that our current measure is not the best at reflecting this, then it makes sense to look elsewhere for a measure of well-being. One such idea is to use something called the Measure of Domestic Progress (MDP). The MDP looks at many of the factors that we might associate with economic growth but takes into consideration the relative effects of economic growth and other things that GDP calculations do not consider. For example, there is an attempt to place a value on the amount of unpaid domestic work that is carried out, which is taken as being a positive factor in contributing to well-being. It also assigns a negative effect to various social and environmental impacts of growth such as pollution, depletion of natural resources, and costs of crime and family breakdown.

Other attempts to suggest more effective measures of improvements in well-being include such things as the Measure of Economic Welfare (MEW) developed by James Tobin and William Nordhaus, the Index of Sustainable Economic Welfare, and the Genuine Progress Indicator.

Personal Well-Being Measures

To take into consideration some of the limitations of GDP and to make use of some of the research carried out on well-being, a number of national statistical offices now publish data which attempt to measure subjective and objective well-being. The ONS in the UK is one such body. It has published reports on personal well-being in the UK since 2012 using four 'high-level' measures and 41 headline measures. The report is produced through a survey which asks around 165,000 people a series of questions. The four 'high-level' measures are:

1. Overall, how satisfied are you with your life nowadays?
2. Overall, to what extent do you feel the things you do in your life are worthwhile?
3. Overall, how happy did you feel yesterday?
4. Overall, how anxious did you feel yesterday?

Answers are given on a scale of 0 to 10 with 10 being 'completely' and 0 being 'not at all'. The headline measures include areas such as personal health and finances, the environment, relationships, where people live, education and skills, the economy and governance. The ONS hopes that the reports offer a further insight into national well-being which supplements other data such as GDP measures. Given the report has only been published since 2012, the ability to identify longer-term trends is in its infancy, but one key factor that has been noted is that there is a relationship between reported well-being and the state of the economy, with unemployment being a particular contributor to low levels of well-being.

Other statistical offices such as Eurostat and the OECD are also developing well-being statistics which reflect the recognition that GDP figures alone do not offer a full picture of well-being. Eurostat's Quality of Life indicators include similar categories to the ONS reports and include data on material living conditions, work, health, education, leisure and social interaction, economic and physical safety, governance and basic rights, the natural and living environment, and the overall experience of life.

INTERNATIONAL DIFFERENCES IN GDP AND THE QUALITY OF LIFE

GDP data is used as a way of comparing well-being across different countries. Rich and poor countries have vastly different levels of GDP per person. Data from the World Bank shows that in 2017, the average GDP per capita in constant 2010 US dollars in high-income countries was around \$41,538 whereas in middle-income countries it was \$4,992. Life expectancy at birth in high-income countries in 2016 is 80, whereas in low-income countries, the figure is 63. Data from UNESCO shows that literacy rates in countries such as Afghanistan were around 32 per cent in 2011, 60 per cent in Bangladesh and 29 per cent in Benin, whereas in developed countries, the rates are now almost 100 per cent.

These data show a clear pattern. In rich countries people can expect to live longer, and almost all of the population can read. In poor countries, people typically can expect to live much shorter lives, and the proportion of the population which is literate is relatively low.

Although data on other aspects of the quality of life are less complete, they tell a similar story. Countries with low GDP per capita tend to have more infants with low birth weight, higher rates of infant mortality, higher rates of maternal mortality, higher rates of child malnutrition and less common access to safe drinking water. In countries with low GDP per capita, fewer school age children are actually in school, and those who are in school must learn with fewer teachers per student. These countries also tend to have fewer televisions, fewer telephones, fewer paved roads and fewer households with electricity.

MEASURING THE COST OF LIVING

When people get their wages and buy the goods and services they need to live and enjoy life, the price of these goods and services determines how many goods and services can be bought. Over time, prices change, and it can be difficult to make comparisons of wages and prices over time. The **cost of living** refers to how much money people need to maintain certain standards of living in terms of the goods and services they can afford to buy. In the UK in 1900, for example, a copy of *The Times* newspaper was priced at 1.2p, a pint of beer or milk could be bought for less than 1p and a dozen fresh eggs was 6.9p. In 1930, the average house price in the UK was £590; a loaf of bread was around 3p (these prices are expressed in decimal currency equivalents). At the turn of the twenty-first century, the average level of prices was 66 times the level of 1900. Because prices were so much lower in the early 1900s than they are today, it is not clear whether people enjoyed a lower or higher standard of living given average wages in 1900 compared to average wages today.

cost of living how much money people need to maintain standards of living in terms of the goods and services they can afford to buy

Taking a snapshot of prices of goods and services in an economy at a particular point in time gives the **price level**. Looking at how the price level changes over time is the rate of change of the price level. To see how the price level changes and, therefore, to provide a way of comparing the cost of living over time, we need to find some way of turning money figures into meaningful measures of purchasing power. That is exactly the job of a statistic called the *Consumer Prices Index* (CPI).

price level a snapshot of the prices of goods and services in an economy at a particular period of time

The CPI is used to monitor changes in the cost of living over time. When the CPI rises, the typical family must spend more money to maintain the same standard of living. Economists use the term *inflation* to describe a situation in which the economy's overall price level is rising and *deflation* when the overall price level is falling. The *inflation rate* is the percentage change in the price level from the previous period; if this is negative, the economy is experiencing deflation.

THE CONSUMER PRICES INDEX

The **Consumer Prices Index** (CPI) is a measure of the overall prices of the goods and services bought by a typical consumer. Each month, a government body (in the UK the ONS and in Europe, Eurostat) computes and reports the CPI.

Consumer Prices Index a measure of the overall prices of the goods and services bought by a typical consumer

How the Consumer Prices Index Is Calculated

To see how these statistics are constructed, let's consider a simple economy in which consumers buy only two goods – salad and burgers. Table 20.3 shows the five steps that the ONS and Eurostat follow. (We will use the ONS as the base for the example here, but the principle applies to the way price changes are measured in Europe as a whole.)

TABLE 20.3**Calculating the Consumer Prices Index and the Inflation Rate:
An Example**

This table shows how to calculate the CPI and the inflation rate for a hypothetical economy in which consumers buy only salad and burgers.

Step 1: Survey consumers to determine a fixed basket of goods		
4 salads and 2 burgers.		
Step 2: Find the price of each good in each year		
Year	Price of salad (€)	Price of burgers (€)
2019	1	2
2020	2	3
2021	3	4
Step 3: Compute the cost of the basket of goods in each year		
2019	(€1 per salad × 4 salads) + (€2 per burger × 2 burgers) = €8	
2020	(€2 per salad × 4 salads) + (€3 per burger × 2 burgers) = €14	
2021	(€3 per salad × 4 salads) + (€4 per burger × 2 burgers) = €20	
Step 4: Choose one year as a base year (2019) and compute the CPI in each year		
2019	$\left(\frac{€8}{€8}\right) \times 100 = 100$	
2020	$\left(\frac{€14}{€8}\right) \times 100 = 175$	
2021	$\left(\frac{€20}{€8}\right) \times 100 = 250$	
Step 5: Use the CPI to compute the inflation rate from previous year		
2019	$\left(\frac{(175 - 100)}{100}\right) \times 100 = 75\%$	
2020	$\left(\frac{(250 - 175)}{175}\right) \times 100 = 43\%$	

- 1. Fix the basket.** The first step in computing the CPI is to determine which prices are most important to the typical consumer. If the typical consumer buys more salad than burgers, then the price of salad is more important than the price of burgers and, therefore, should be given greater weight in measuring the cost of living. The ONS sets these weights by surveying consumers and finding the basket of goods and services that the typical consumer buys. In the example in Table 20.3, the typical consumer buys a basket of four salads and two burgers.
- 2. Find the prices.** The second step is to find the prices of each of the goods and services in the basket for each point in time. The table shows the prices of salad and burgers for three different years.
- 3. Compute the basket's cost.** The third step is to use the data on prices to calculate the cost of the basket of goods and services at different times. The table shows this calculation for each of the three years. Notice that only the prices in this calculation change. By keeping the basket of goods the same (four salads and two burgers), we are isolating the effects of price changes from the effect of any quantity changes that might be occurring at the same time.
- 4. Choose a base year and compute the index.** The fourth step is to designate one year as the base year, which is the benchmark against which other years are compared. To calculate the index, the price of the basket of goods and services in each year is divided by the price of the basket in the base year, and this ratio is then multiplied by 100. The resulting number is the CPI.

In the example in Table 20.3, the year 2019 is the base year. In this year, the basket of salad and burgers costs €8. Therefore the price of the basket in all years is divided by €8 and multiplied by 100. The CPI is 100 in 2019. (The index is always 100 in the base year.) The CPI is 175 in 2020. This means that

the price of the basket in 2020 is 175 per cent of its price in the base year. Put differently, a basket of goods that costs €100 in the base year costs €175 in 2020. Similarly, the CPI is 250 in 2021, indicating that the price level in 2021 is 250 per cent of the price level in the base year.

- 5. Compute the inflation rate.** The fifth and final step is to use the CPI to calculate the **inflation rate**, which is the percentage change in the price index from the preceding period. That is, the inflation rate between two consecutive years is computed as follows:

$$\text{Inflation rate in year 2} = 100 \times \frac{(\text{CPI in year 2} - \text{CPI in year 1})}{\text{CPI in year 1}}$$

In our example, the inflation rate is 75 per cent in 2020 and 43 per cent in 2021.

$$\text{CPI}_{\text{year } t} = \frac{\text{Cost of basket of goods in year } t \text{ prices}}{\text{Cost of basket of goods in base year prices}}$$

The inflation rate is given by:

$$\text{Inflation}_{\text{year } t} = \frac{(\text{Price index}_{\text{year } t}) - (\text{Price index}_{\text{year } t-1})}{(\text{Price index}_{\text{year } t-1})} \times 100$$

inflation rate the percentage change in the price index from the preceding period

Although this example simplifies the real world by including only two goods, it shows how statistics offices compute the CPI and the inflation rate. The offices collect and process data on the prices of thousands of goods and services every month and, by following the five foregoing steps, determines how quickly the cost of living for the typical consumer is rising. In addition to the CPI for the overall economy, statistics offices also calculate price indices for the sub-categories of ‘goods’ and of ‘services’ separately, as well as the **producer prices index** (PPI), which measures the prices of a basket of goods and services bought by firms rather than consumers. Because firms eventually pass on their costs to consumers in the form of higher consumer prices, changes in the PPI are often thought to be useful in predicting changes in the CPI.

producer prices index a measure of the prices of a basket of goods and services bought by firms

Problems in Measuring the Cost of Living

The goal of the CPI is to measure changes in the cost of living – how much people must pay to purchase goods and services. In other words, the CPI tries to gauge how much incomes must rise to maintain a constant standard of living. Assume that an individual in 2019 has an income of €120 per week and only buys burgers at €2 each. Their standard of living is 60 burgers a week. In 2020, burgers rise in price to €3 and so the standard of living, assuming income does not change, is now only 40 burgers a week. We would say that the individual's standard of living has fallen because they can now afford to consume fewer burgers. To keep their standard of living constant at 60 burgers, the person's income needs to rise from €120 per week to €180 per week. If the rise in incomes keeps pace with the rise in prices, then the individual's standard of living will remain constant. If incomes rise by a lower percentage than the CPI, then standards of living are being eroded. If incomes rise at a faster rate than the CPI, standards of living are improving, and people are better off.

The CPI, however, is not a perfect measure of the cost of living. Three problems with the index are widely acknowledged but are difficult to solve.

Substitution Bias The first problem is called *substitution bias*. When prices change from one year to the next, they do not all change proportionately: some prices rise more than others and some prices fall. Consumers respond to these differing price changes by buying less of the goods whose prices have risen

by large amounts and buying more of the goods whose prices have risen less or perhaps even have fallen (and, of course, different consumers will respond to price changes in different ways because of the price elasticity of demand). Consumers substitute towards goods that have become relatively less expensive. If a price index is computed assuming a fixed basket of goods, it ignores the possibility of consumer substitution and, therefore, overstates the increase in the cost of living from one year to the next.

Let's consider a simple example. Imagine that in the base year apples are cheaper than pears, and so consumers buy more apples than pears. When the statistics office constructs the basket of goods, apples will be weighted more heavily than pears. Suppose that next year pears are cheaper than apples. Consumers may respond to the price changes by buying more pears and fewer apples. Yet, when computing the CPI, the statistics office uses a fixed basket, which in essence assumes that consumers continue buying the now expensive apples in the same quantities as before. For this reason, the index will measure a much larger increase in the cost of living than consumers actually experience.

The Introduction of New Goods The second problem with the CPI is the *introduction of new goods*. When a new good is introduced, consumers have more variety from which to choose. Greater variety, in turn, makes each euro or pound more valuable, so consumers need fewer units of currency to maintain any given standard of living. Yet because the CPI is based on a fixed basket of goods and services, it does not reflect this change in purchasing power.

For example, when video players were introduced, consumers were able to watch their favourite films at home. Compared with going to the cinema, the convenience was greater and the cost was less. A perfect cost of living index would have reflected the introduction of the video player with a decrease in the cost of living. The CPI, however, did not decrease in response to the introduction of the video player. Eventually, the ONS did revise the basket of goods to include video players, and subsequently the index reflected changes in their prices. As time has passed the index has had to be revised to take account of the decline in the video and the rise of the DVD, then Blu-ray DVDs and, more recently, the increase in downloading and streaming of films.

Unmeasured Quality Change The third problem with the CPI is *unmeasured quality change*. If the quality of a good deteriorates from one year to the next, the effective value of a euro or pound falls, even if the price of the good stays the same. Similarly, if the quality rises from one year to the next, the effective value of a euro or pound rises. Statistics offices do their best to account for quality change. When the quality of a good in the basket changes – for example, when a car model has more horsepower or gets better fuel mileage from one year to the next – statistics offices adjust the price of the good to account for the quality change. They are, in essence, trying to compute the price of a basket of goods of constant quality.

To take another example, in 2004 the ONS introduced digital cameras into the CPI for the first time. As well as the problems associated with introducing a new good into the index that we discussed earlier, digital cameras are also subject to very rapid technological progress – features such as zoom and the number of megapixels keep on improving rapidly, and the sale of digital cameras has now been impacted by the developments in camera technology which is incorporated into smartphones. Thus while the average price of a digital camera or a smartphone might remain the same over a period, the average quality may have risen substantially. The ONS attempts to correct for this by a method known as *hedonic quality adjustment*. This involves working out the average characteristics (e.g. screen size, number of megapixels, zoom features, etc.) of the average digital camera and adjusting the price when one of these average characteristics increases. Despite these efforts, changes in quality remain a problem, because quality is so hard to measure.

Relevance of the CPI A final problem with the index is that people may not see the reported CPI measure of inflation as relevant to their particular situation. This is because their spending patterns are individual and might not be typical of the representative pattern on which the official figures are based. For example, if an individual spent a high proportion of their income on fuel and their mortgage, the effect of price rises in gas, electricity, petrol, and a rise in mortgage rates would have a disproportionate effect on their experience of inflation. This different perception of inflation can have an effect on expectations, the importance of which we will see later.

Because of the different perceptions that people have about inflation and how it affects them, the ONS published a Personal Inflation Calculator (PIC) in 2007. The PIC allows users to be able to input their own details such as what their personal monthly expenditure is, how much they spend on food, meals out, alcohol, clothing and footwear, fuel for transport and so on. In addition, the calculator looks at what is spent on utilities such as water, council tax, vehicles, holidays and housing.

The ONS hopes that the PIC will help people to develop more of an interest in how inflation is calculated and be more aware of how the reality of price rises in their lives might differ to the officially published figures.

FYI

What Is in the CPI's Basket?

When constructing the CPI, the UK ONS tries to include all the goods and services that the typical consumer buys. Moreover, it tries to weight these goods and services according to how much consumers buy of each item. Every month the ONS collects around 180,000 prices of 700 goods and services from around 150 areas throughout the UK that are supposed to be representative of the goods and services used on a regular basis. As buying habits change, the basket of goods and services must change also. Each year the ONS announces a revision to the basket of goods and services that make up the basis of inflation figures in the UK. The ONS must take into consideration not only how representative the sample of goods and services is but also any changes in importance in the typical household budget, and whether the replacement brands that shops bring in are of comparable quality.

Some goods and services that are not as representative any more are removed from the basket. This is measured by the amount spent on these goods and services. If consumer expenditure on items exceeds £400 million annually, then these are normally included in the basket, but if expenditure falls below £100 million, then there must be a very good reason put forward to keep these goods in the basket. In recent years this has included microwave ovens, which first made an appearance in the index in the 1980s. They are still popular, say the ONS, but their reliability and falling prices means that the amount spent on them has reduced, and as a result they have been removed from the basket. Other items that have dropped out include film for 35 mm cameras (the cameras themselves dropped out in 2007), top 40 CD singles, and television repairs. As technology changes the way we live, the ONS says that people prefer to download music rather than buy singles, and households tend to replace TVs rather than get them repaired.

The goods and services that have come into the basket in recent years include e-cigarette refills and liquid, craft beers, protein powder, games consoles, online subscriptions and music streaming subscription.

The GDP Deflator versus the Consumer Prices Index

Recall that the GDP deflator is the ratio of nominal GDP to real GDP. Because nominal GDP is current output valued at current prices, and real GDP is current output valued at base year prices, the GDP deflator reflects the current level of prices relative to the level of prices in the base year.

Economists and policymakers monitor both the GDP deflator and the CPI to gauge how quickly prices are rising. Usually, these two statistics tell a similar story. Yet there are two important differences that can cause them to diverge.

The first difference is that the GDP deflator reflects the prices of all goods and services *produced domestically*, whereas the CPI reflects the prices of all goods and services *bought by consumers*. For example, suppose that the price of an aeroplane produced by Dassault, a French aerospace firm, and sold to the French Air Force rises. Even though the aeroplane is part of GDP in France, it is not part of the basket of goods and services bought by a typical consumer. Thus the price increase shows up in the GDP deflator for France but not in France's CPI.

As another example, suppose that Volvo raises the price of its cars. Because Volvos are made in Sweden, the car is not part of French GDP. But French consumers buy Volvos, and so the car is part of the

typical consumer's basket of goods. Hence a price increase in an imported consumption good, such as a Volvo, shows up in the CPI but not in the GDP deflator.

This first difference between the CPI and the GDP deflator is particularly important when the price of oil changes. Although the UK does produce some oil, as with many parts of Europe, much of the oil used in the UK is imported from the Middle East. As a result, oil and oil products such as petrol and heating oil comprise a much larger share of consumer spending than they do of GDP. When the price of oil changes, the CPI changes by much more than does the GDP deflator.

The second and subtler difference between the GDP deflator and the CPI concerns how various prices are weighted to yield a single number for the overall level of prices. The CPI compares the price of a *fixed* basket of goods and services with the price of the basket in the base year. While, as we have seen, the ONS revises the basket of goods on a regular basis, in contrast, the GDP deflator compares the prices of *currently produced* goods and services with the prices of the same goods and services in the base year. Thus the group of goods and services used to compute the GDP deflator changes automatically over time. This difference is not important when all prices are changing proportionately. However, if the prices of different goods and services are changing by varying amounts, the way we weight the various prices matters for the overall inflation rate.

The inflation rate as measured by both the GDP deflator and the CPI do tend to move together.

SELF TEST Explain briefly what the CPI is trying to measure and how it is constructed.

CORRECTING ECONOMIC VARIABLES FOR THE EFFECTS OF INFLATION

The purpose of measuring the overall level of prices in the economy is to permit comparisons of monetary figures from different points in time. Now that we know how price indices are calculated, let's see how we might use such an index to compare a certain figure from the past to a figure in the present.

Money Figures from Different Times

To compare money figures over a period of time, we need to know the level of prices in both the historical year and the level of prices today. To compare money figures, we need to inflate the historical figure into today's currency. A price index determines the size of this inflation correction.

The formula for turning euro or pound figures from *year T* into today's euros or pounds is the following (assuming the price level today and the price level in *year T* are measured against the same base year):

$$\text{Amount in today's currency} = \text{Amount in year } T \text{ currency} \times \frac{\text{Price level today}}{\text{Price level in year } T}$$

A price index such as the CPI measures the price level and determines the size of the inflation correction. To illustrate, let us take an example of the salary for members of parliament in the UK. In 1947 this was £1,000 per year according to the House of Commons Information Office. The ONS reports the prices index (PI) for 1947 at 28.9 and for 2020, 1018.6. Substituting these figures into the formula gives:

$$\text{Salary in 2020 } \pounds = \text{Salary in 1947 } \pounds \times \frac{\text{PI in 2020}}{\text{PI in 1947}}$$

$$\text{Salary in 2020} = \pounds 1000 \times \left(\frac{1018.6}{28.9} \right)$$

$$\text{Salary in 2020} = \pounds 35,230$$

We find that the salary paid to an MP in 1947 is equivalent to a salary in 2020 of £35,230. Assume that the basic annual salary of an MP in 2020 was £74,000. It seems that an MP in 2020 is considerably better off in terms of salary earned than their compatriots in 1947.

Indexation

As we have just seen, price indices are used to correct for the effects of inflation when comparing monetary figures from different times. This type of correction shows up in many places in the economy. When some money amount is automatically corrected for inflation by law or contract, the amount is said to be **indexed** for inflation. For example, many long-term contracts between firms and unions include partial or complete indexation of the wage to the CPI. Such a provision is called a *cost of living allowance*, or COLA. A COLA automatically raises the wage each year based on the CPI or other measure, at a particular point in time.

indexed the automatic correction of a money amount for the effects of inflation by law or contract

Income tax brackets – the income levels at which the tax rates change – are also often moved annually in line with inflation, although, in most countries, they are not formally indexed. Indeed, there are many ways in which the tax system is not indexed for inflation, even when perhaps it should be.

Real and Nominal Interest Rates

Correcting economic variables for the effects of inflation is particularly important, and somewhat tricky, when we look at data on interest rates. Savings deposited in an interest-bearing account provide a return based on the interest rate available. Conversely, when people borrow from a bank to fund purchases such as a car, interest is payable on the loan. Interest represents a payment in the future for a transfer of money in the past. As a result, interest rates always involve comparing amounts of money at different points in time. To fully understand interest rates, we need to know how to correct for the effects of inflation.

Let's consider an example. Suppose that Carla deposits €1,000 in a savings account that pays an annual interest rate of 10 per cent. After a year passes, Carla has accumulated €100 in interest. Carla then withdraws her €1,100. Is Carla €100 richer than she was when she made the deposit a year earlier?

The answer depends on what we mean by 'richer'. Carla does have €100 more than she had before. In other words, the number of euros has risen by 10 per cent. If prices have risen at the same time, each euro now buys less than it did a year ago. Thus her purchasing power has not risen by 10 per cent. If the inflation rate was 4 per cent, then the amount of goods she can buy has increased by only 6 per cent. If the inflation rate was 15 per cent, then the price of goods has increased proportionately more than the number of euros in her account. In that case, Carla's purchasing power has actually fallen by 5 per cent.

The interest rate that the savings account pays is called the **nominal interest rate**, and the interest rate corrected for inflation is called the **real interest rate**. We can write the relationship between the nominal interest rate, the real interest rate and inflation as follows:

$$r_t = i_t - \pi_t$$

The real interest rate in a particular time period (r_t) = nominal interest rate in that time period (i_t) – inflation rate in the same time period (π_t).

The real interest rate is the difference between the nominal interest rate and the rate of inflation. The nominal interest rate tells you how fast the number of pounds or euros in your bank account rises over time. The real interest rate tells you how fast the purchasing power of your bank account rises over time.

nominal interest rate the interest rate as usually reported without a correction for the effects of inflation
real interest rate the interest rate corrected for the effects of inflation

SELF TEST Why is it important to adjust for inflation when considering wage increases and the returns from any financial investment?

CONCLUSION

This chapter has discussed how economists measure well-being and the cost of living. Much of the study of macroeconomics we will look at in the rest of this book is aimed at revealing the long-run and short-run determinants of a nation's GDP. Why, for example, do most economies experience fluctuations in economic activity? Why are key macroeconomic measures such as inflation, unemployment, interest rates and exchange rates important in determining economic well-being? What can governments do to promote more rapid growth in GDP? What can policymakers do to reduce the severity of fluctuations in GDP? These are the questions we will take up shortly.

Despite the focus on GDP as a measure of economic well-being, we must not forget the limitations of this measure and that overall economic well-being may include many factors other than income.

When comparing figures in monetary or nominal terms from different times, it is important to keep in mind that a pound or euro today is not the same as a pound or euro 10 years ago or, most likely, 10 years from now. We have seen how economists measure the overall level of prices in the economy and how they use price indices to correct economic variables for the effects of inflation. This analysis is only a starting point. We have not yet examined the causes and effects of inflation or how inflation interacts with other economic variables. To do that, we need to go beyond issues of measurement. Having explained how economists measure macroeconomic quantities and prices, we are now ready to develop the models that explain long-run and short-run movements in these variables.

SUMMARY

- Because every transaction has a buyer and a seller, the total expenditure in the economy must equal the total income in the economy.
- GDP measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services. More precisely, GDP is the market value of all final goods and services produced within a country in a given period of time.
- GDP is divided among four components of expenditure: consumption, investment, government purchases and net exports. Consumption includes spending on goods and services by households, with the exception of purchases of new housing. Investment includes spending on new equipment and structures, including households' purchases of new housing. Government spending includes spending on goods and services by local and central governments. Net exports equal the value of goods and services produced domestically and sold abroad (exports) minus the value of goods and services produced abroad and sold domestically (imports).
- Nominal GDP uses current prices to value the economy's production of goods and services. Real GDP uses constant base year prices to value the economy's production of goods and services. The GDP deflator – calculated from the ratio of nominal to real GDP – measures the level of prices in the economy.
- GDP is one measure of economic well-being. It has limitations because it excludes many activities which occur illegally and does not take account of the benefits to well-being from work done outside markets such as looking after the home, rearing children, caring and volunteer work.
- Because of the limitations of using GDP, a number of governments are now supporting efforts to report on broader measures of well-being which include both subjective and objective reporting of how people feel about their lives and circumstances.
- The CPI shows the changes in the prices of a basket of goods and services relative to the prices of the same basket in the base year. The index is used to measure the overall level of prices in the economy. The percentage change in the CPI measures the inflation rate.
- The CPI is an imperfect measure of the cost of living for three reasons. First, it does not take into account consumers' ability to substitute towards goods that become relatively cheaper over time. Second, it does not take into account increases in the purchasing power of money due to the introduction of new goods. Third, it is distorted by unmeasured changes in the quality of goods and services. Because of these measurement problems, the CPI overstates true inflation.
- Although the GDP deflator also measures the overall level of prices in the economy, it differs from the CPI because it includes goods and services produced rather than goods and services consumed. As a result, imported goods

affect the CPI but not the GDP deflator. In addition, while the CPI uses a fixed basket of goods, the GDP deflator automatically changes the group of goods and services over time as the composition of GDP changes.

- Money figures (e.g. in euros) from different points in time do not represent a valid comparison of purchasing power. To compare a money figure from the past to a money figure today, the older figure should be inflated using a price index.
- Various laws and private contracts use price indices to correct for the effects of inflation.
- A correction for inflation is especially important when looking at data on interest rates. The nominal interest rate is the interest rate usually reported; it is the rate at which the amount of money in a savings account increases over time. In contrast, the real interest rate takes into account changes in the value of the money over time. The real interest rate equals the nominal interest rate minus the rate of inflation.

IN THE NEWS

Personal Well-Being in the UK

The ONS in the UK produces a report called *Personal Well-Being in the UK*. This In the News looks at some of the highlights from the report published in 2018, for the period June 2017 to June 2018.

The ONS report showed that there was no change in average life satisfaction in the UK. Happiness and anxiety ratings did not change, but fewer people reported low happiness ratings and there were more people who said their anxiety rating was 'low', with much of this improvement being driven by ratings from England. When looking at the different regions of the UK, reports of personal well-being ratings in Wales were lower than in the rest of the UK, better in Northern Ireland and no change in Scotland.

Of the four high-level measures, satisfaction with life was 7.7, feeling that what one does in life is worthwhile at 7.9 and happiness yesterday, 7.5. The report on feelings of anxiety yesterday was 2.9. All these figures are out of 10. The report showed that compared to the years ending June 2017 and June 2018, there were no significant changes in any of the key measures of personal well-being in the UK.

The report offered some tentative reasons for the findings. The improvement in the economy since the Financial Crisis 2007–9 may be one reason and in particular the improvement in the labour market where unemployment was at historically low levels.

There were geographical differences highlighted by the report. Annual gross disposable household income in Wales was some £3,600 lower than in England at £15,835. Needless to say, reports of being mostly or completely satisfied with their income was lower at 42.1 per cent compared to 45.6 per cent in England. The median full-time weekly wage for Wales was second to the North East of England.

Critical Thinking Questions

- 1 What do you think is the value of the *Personal Well-Being* report and why do you think the ONS produces it?
- 2 What do you think is a better measure of well-being, a self-report survey style measure such as the *Personal Well-Being* report or GDP?



Well-being is not just measured by changes in GDP; self-reporting of how people feel about themselves and their lives has become an important measure of well-being.

(Continued)

- 3 Of the four high-level measures, would you expect that if the first three show increased reports of well-being that the fourth must always be falling? Explain.
- 4 Inflation rose faster than wages in the UK for several years between 2010 and 2017. How do you think this might affect reports in the four high-level measures?
- 5 How reliable do you think reports such as the *Personal Well-Being* report are as a measure of well-being in the economy?

Reference: *Personal Well-Being in the UK July 2017 to June 2018*. Office for National Statistics. www.ons.gov.uk/peoplepopulationandcommunity/wellbeing/bulletins/measuringnationalwellbeing/july2017tojune2018/pdf, accessed 28 December 2018.

QUESTIONS FOR REVIEW

- 1 Explain why an economy's income must equal its expenditure.
- 2 What is meant by the term GDP per capita and how is it measured?
- 3 List the four components of GDP. Give an example of each.
- 4 Why do economists use real GDP rather than nominal GDP to gauge economic well-being?
- 5 In the year 2020, the economy produces 100 loaves of bread that sell for €2 each. In the year 2021, the economy produces 200 loaves of bread that sell for €3 each. Calculate nominal GDP, real GDP and the GDP deflator for each year (use 2020 as the base year). By what percentage does each of these three statistics rise from one year to the next?
- 6 What are the limitations of using GDP as a measure of well-being for a country?
- 7 What are the five stages of constructing a prices index?
- 8 Describe the three problems that make the CPI an imperfect measure of the cost of living.
- 9 Assume that the price of a bottle of wine in 1990 was €3.50 and in 2021 it is €8.50. Further assume that the price index in 1990 was 95 and in 2021 was 160. Was wine cheaper in 1990 than in 2021? Explain.
- 10 Explain the meaning of nominal interest rate and real interest rate. How are they related?

PROBLEMS AND APPLICATIONS

- 1 What components of GDP (if any) would each of the following transactions affect? Explain.
 - a. A family buys a new refrigerator.
 - b. Aunt Jane buys a new house.
 - c. Aston Martin sells a DB7 from its inventory.
 - d. You buy a pizza.
 - e. The government builds a new motorway.
 - f. You buy a bottle of Californian wine.
 - g. Honda expands its factory in Derby, England.
- 2 Why do you think households' purchases of new housing are included in the investment component of GDP rather than the consumption component? Can you think of a reason why households' purchases of new cars should also be included in investment rather than in consumption? To what other consumption goods might this logic apply?
- 3 Below are some data from the land of milk and honey.

Year	Price of milk (€)	Quantity of milk (litres)	Price of honey (€)	Quantity of honey (jars)
2019	1	100	2	50
2020	1	200	2	100
2021	2	200	4	100

- a. Compute nominal GDP, real GDP and the GDP deflator for each year, using 2019 as the base year.
 - b. Compute the percentage change in nominal GDP, real GDP and the GDP deflator in 2020 and 2021 from the preceding year. For each year, identify the variable that does not change. Explain in words why your answer makes sense.
 - c. Did economic well-being rise more in 2020 or 2021? Explain.
- 4 Economists sometimes prefer to use GNP rather than GDP as a measure of economic well-being. Which measure should we prefer if we are analyzing the total income of domestic residents? Which measure should we prefer if we are analyzing the total amount of economic activity occurring in the economy?
 - 5 Given the limitations of using GDP as a measure of well-being, should calculation of GDP be scrapped in favour of wider reporting of well-being such as objective and subjective well-being?
 - 6 The participation of women in many European and North American economies has risen dramatically since the 1990s.
 - a. How do you think this rise affected GDP?
 - b. Now imagine a measure of well-being that includes time spent working in the home and taking leisure. How would the change in this measure of well-being compare to the change in GDP?
 - c. Can you think of other aspects of well-being that are associated with the rise in women's labour force participation? Would it be practical to construct a measure of well-being that includes these aspects?
 - 7 Suppose that people consume only three goods, as shown in this table:

	Tennis balls	Tennis racquets	Cola
2019 price (€)	2	40	1
2019 quantity	100	10	200
2020 price (€)	2	60	2
2020 quantity	100	10	200

- a. What is the percentage change in the price of each of the three goods? What is the percentage change in the overall price level?
 - b. Have tennis racquets become more or less expensive relative to cola? Does the well-being of some people change relative to that of others? Explain.
- 8 Which of the problems in the construction of the CPI might be illustrated by each of the following situations? Explain.
 - a. The increase in streaming films.
 - b. The introduction of air bags in cars.
 - c. Increased personal computer purchases in response to a decline in their price.
 - d. Increased use of digital cameras in smartphones.
 - e. Reduced use of fuel-efficient cars after fuel prices fall.
 - 9 Suppose the government were to determine the level of the state retirement pension in the UK so that it increased each year in proportion to the increase in the CPI.
 - a. If the elderly consume the same market basket as other people, would such a policy provide the elderly with an improvement in their standard of living each year? Explain.
 - b. In fact, the elderly consume more healthcare than younger people, and healthcare costs tend to rise faster than overall inflation. What would you do to determine whether the elderly are actually better off from year to year?
 - 10 Suppose that a borrower and a lender agree on the nominal interest rate to be paid on a loan. Then inflation turns out to be higher than they both expected.
 - a. Is the real interest rate on this loan higher or lower than expected?
 - b. Does the lender gain or lose from this unexpectedly high inflation? Does the borrower gain or lose?

